

Carla Becker

PROGRAM & TECHNICAL LEADERSHIP

- **Program & Execution**

0-to-1 program development; technical program management, roadmap definition, early-stage project ramp-up; cross-functional coordination across ML, software, hardware, and operations; vendor management, SOWs, budgeting, and delivery against milestones.

- **AI/ML & Systems**

Machine learning model evaluation and lifecycle management; experimentation pipelines, data quality, and deployment tradeoffs under compute and latency constraints; simulation-ML hybrid systems (physics-based + learned models).

- **Technical Stack**

Python, SQL, C++, Flask, Snowflake, Docker/Kubernetes, Unix/Bash; data visualization, internal tooling, automation, and web apps.

PROFESSIONAL EXPERIENCE

- **Founding Program Manager & Product Engineer**, Sakuu Corp.

May 2018 - June 2022

- Third engineer hired at early-stage solid-state battery startup; helped scale company from 10 to 60 people while building core R&D programs in a pre-product-market-fit environment without established playbooks.
- Led a \$300K, two-year program to co-develop a browser-based ML and analytics platform with an external software partner, managing stakeholders across two companies and serving as product owner for requirements and UI/UX.
- Built a modeling & simulation business unit from scratch: defined 3-year roadmap, scope, and success metrics; owned vendor selection, SOWs, and alignment across engineering, operations, and leadership.
- Acted as connective tissue between R&D strategy and daily execution: partnered with staff engineers on direction, then coordinated a 10-person team to deliver against shifting priorities and investor milestones.

- **Technical Lead, Design for Manufacturing**, Apple Inc.

February 2025 - August 2025

- Owned end-to-end development of an internal AI-enabled platform for component-level data ingestion, visualization, and live editing, used across DFM workflows.
- Coordinated ML modeling, backend infrastructure (Snowflake, Flask), deployment (Docker/K8s), and user feedback loops.
- Led evaluation of 18 ML approaches for solder joint failure prediction, framing tradeoffs between accuracy, interpretability, and compute cost to equip leadership with actionable recommendations.
- Proposed a novel hybrid FEA + neural-network approach to reduce system-level simulation time by orders of magnitude and road-mapped its development.

- **Program Manager & Technical Lead**, UC Berkeley

June 2022 - Present

- **Program Manager, DEWA:** Owned execution of an international engineering certificate program administered jointly by UC Berkeley and Dubai Electricity & Water Authority (DEWA). Coordinated faculty, registrars, TAs, and DEWA leadership across time zones.
- **Technical Lead, Capstone Connect:** Led design and delivery of an automated matching platform supporting hundreds of industry-sponsored MEng capstone projects. Translated multi-stakeholder requirements into ranking logic and system constraints.

RESEARCH EXPERIENCE

- **Graduate Student Researcher**, UC Berkeley

May 2021 - Present

- Led development of ML-driven optimization and simulation tools across materials discovery ([The Materials Project](#)) and agricultural systems ([AI Institute for Next-Gen Food Systems](#)), spanning ODE models, ray tracing, and scalable experimentation pipelines.
- Experience translating research prototypes into usable internal tools and platforms (APIs, AWS, testing, performance optimization in Python).
- Outstanding Graduate Student Instructor 2024, Modeling and Simulation of Advanced Manufacturing Systems.

CONTACT

Linkedin: [linkedin.com/in/carla-becker](https://www.linkedin.com/in/carla-becker)

Email: carla.joy.becker@gmail.com

GitHub: github.com/carlacupcake

EDUCATION

- **University of California Berkeley**

Ph.D., Mechanical Engineering;

Expected: May 2026

M.S., Mechanical Engineering, 2024

M.S., Materials Science & Engineering, 2022

- **Harvey Mudd College**

B.S. Physics, B.S. Chemistry, 2018